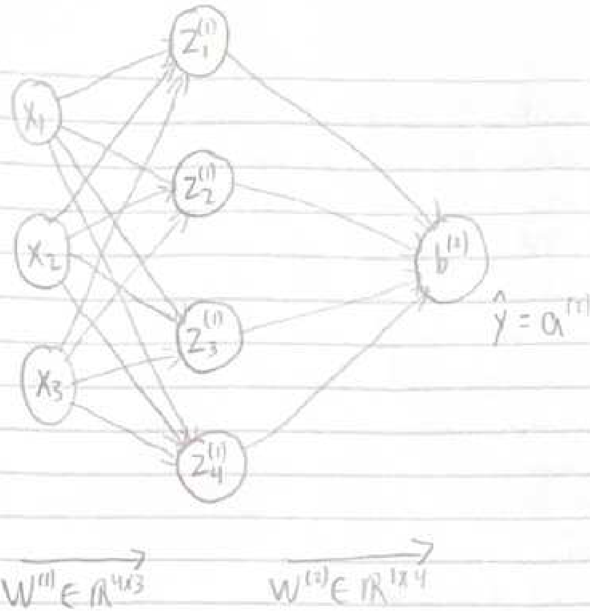
Activation:

$$h(a^{(1)}) = z_i^{(1)} = \tanh(a_i^{(1)})$$



Dimension Verification Table:

Operation	Left operand	Right operand	Result	Verification
$a^{(1)} = W^{(1)}X$	$W^{(1)} \in \mathbb{R}^{4\times3}$	$X \in \mathbb{R}^{3\times1}$	$a^{(1)} \in \mathbb{R}^{4\times1}$	$(4\times3)(3\times1) \rightarrow 4\times1$
$z^{(1)} = h(a^{(1)})$	$a^{(1)} \in \mathbb{R}^{4\times1}$	element-wise	$z^{(1)} \in \mathbb{R}^{4\times1}$	element-wise $\rightarrow$ shape kept
$a^{(1)} = W^{(2)}z^{(1)}$	$W^{(2)} \in \mathbb{R}^{1\times4}$	$z^{(1)} \in \mathbb{R}^{4\times1}$	$a^{(2)} \in \mathbb{R}^{1\times1}$	$(1\times4)(4\times1) \rightarrow 1\times1$

Weight initialization

- Random seed: 20

$$W^{(1)} = \begin{bmatrix} 0.0191 & 0.4947 & 0.0718 \\ 0.3751 & -0.2048 & 0.7237 \\ -0.2023 & 0.5745 & -0.4695 \\ -0.9507 & 0.5477 & -0.1668 \end{bmatrix}$$

- $W^{(1)} \in \mathbb{R}^{4\times3}$ $\quad W^{(1)}: 4\times3$
- $b^{(1)} \in \mathbb{R}^{4\times1}$ $\quad b^{(1)}: 4\times1$
- $W^{(2)} \in \mathbb{R}^{1\times4}$ $\quad W^{(2)}: 1\times4$
- $b^{(2)} \in \mathbb{R}$ $\quad b^{(2)}: 1\times1$

$$b^{(1)} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$W^{(2)} = \begin{bmatrix} -0.7209 & -0.7083 & -0.8148 & -0.9076 \end{bmatrix}$$

$$b^{(2)} = [0]$$

Justification for hidden layer size:

I chose 4 hidden neurons because 3 input features is not very large so a smaller hidden layer is enough. The relationship between sepal length, sepal width and petal width is not a complex task, so a larger hidden layer may introduce unneeded parameters. 4 neurons is enough for the model to be flexible with nonlinear patterns and lessen the risk for overfitting.

Forward Pass Calculations

data point: $x = [5.1, 3.5, 0.2]^T$, $y = 1.4$

Hidden layer pre-activations: compute $a^{(1)} = W^{(1)}x + b^{(1)}$

$$W^{(1)}x = \begin{bmatrix} 0.0191 & 0.4947 & 0.0716 \\ 0.3751 & -0.2048 & 0.7237 \\ -0.2023 & 0.5745 & -0.4695 \\ -0.9507 & 0.5477 & -0.1668 \end{bmatrix} \begin{bmatrix} 5.1 \\ 3.5 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 5.1(0.0191) + 3.5(0.4947) + 0.2(0.0718) \\ 5.1(0.3751) + 3.5(-0.2048) + 0.2(0.7237) \\ 5.1(-0.2023) + 3.5(0.5745) + 0.2(-0.4695) \\ 5.1(-0.9507) + 3.5(0.5477) + 0.2(-0.1668) \end{bmatrix}$$

$$= \begin{bmatrix} 1.8432 \\ 1.3410 \\ 0.8851 \\ -2.9650 \end{bmatrix}$$

$a_1^{(1)} = 0.0191(5.1) + 0.4947(3.5) + 0.0718(0.2) + 0 = 1.8432$

$a_2^{(1)} = 0.3751(5.1) - 0.2048(3.5) + 0.7237(0.2) + 0 = 1.3410$

$a_3^{(1)} = -0.2023(5.1) + 0.5745(3.5) - 0.4695(0.2) + 0 = 0.8851$

$a_4^{(1)} = -0.9507(5.1) + 0.5477(3.5) - 0.1668(0.2) + 0 = -2.9650$

$a^{(1)} = [1.8432 \ 1.3410 \ 0.8851 \ -2.9650]^T$

Each $a_i^{(1)}$ is the pre-activation for each i-th hidden neuron, which is represented by the weighted sum of the input features with the bias.

Hidden layer activations: Compute $z^{(1)} = \tanh(a^{(1)})$

$$z_1^{(1)} = \tanh(1.8432) = \frac{e^{1.8432} - e^{-1.8432}}{e^{1.8432} + e^{-1.8432}} = \frac{6.15841}{6.47503} = 0.9511$$

$$z_2^{(1)} = \tanh(1.3410) = \frac{e^{1.3410} - e^{-1.3410}}{e^{1.3410} + e^{-1.3410}} = \frac{3.56128}{4.08445} = 0.872$$

$$z_3^{(1)} = \tanh(0.8851) = \frac{e^{0.8851} - e^{-0.8851}}{e^{0.8851} + e^{-0.8851}} = \frac{2.01055}{2.83590} = 0.709$$

$$z_4^{(1)} = \tanh(-2.9650) = \frac{e^{-2.9650} - e^{2.9650}}{e^{-2.9650} + e^{2.9650}} = \frac{-19.34314}{19.44626} = -0.9947$$

$$z^{(1)} = [0.9511 \; 0.872 \; 0.709 \; -0.9947]^T$$

Each $z_i^{(1)}$ is the activated output of the i-th hidden neuron that is acquired by using the tanh function to its pre-activation $a_i^{(1)}$. The activation puts the neuron's response range into $(-1, 1)$, which introduces nonlinearity to the network.

Output layer computation: Compute $a^{(2)} = W^{(2)} z^{(1)} + b^{(2)}$ and $\hat{y} = a^{(2)}$

$$W^{(2)} z^{(1)} = \begin{bmatrix} -0.7209 & -0.7083 & -0.8148 & -0.9076 \end{bmatrix} \begin{bmatrix} 0.9511 \\ 0.872 \\ 0.709 \\ -0.9947 \end{bmatrix}$$

$$= \begin{bmatrix} -0.7209(0.9511) - 0.7083(0.872) - 0.8146(0.709) - 0.9076(-0.9947) \end{bmatrix}$$

$$= \begin{bmatrix} -0.9782 \end{bmatrix}$$

$$a^{(2)} = -0.7209(0.9511) + (-0.7083(0.872)) + (-0.8146(0.709)) + (-0.9076(-0.9947)) + 0$$
$$= -0.68565 - 0.61764 - 0.577693 + 0.9027897$$
$$= -0.9782$$

$$\hat{y} = -0.9782$$

$\hat{y}$ is the network's prediction for petal length. It is a weighted linear combination of the hidden layer outputs $z^{(1)}$ with the output bias. Since the activation is linear, the network's prediction is equal to the weighted sum.

Loss Computation: compute $E = \frac{1}{2}(\hat{y} - y)^2$

Prediction error: $\hat{y} - y = -0.9782 - 1.400 = -2.3782$

Squared error: $(\hat{y} - y)^2 = (-2.3782)^2 = 5.6558$

Loss: $E = \frac{1}{2}(\hat{y} - y)^2 = \frac{1}{2}(5.6558) = 2.8279$

The loss equation is $E = \frac{1}{2}(\hat{y} - y)^2$. The prediction error $\hat{y} - y$ tells you how far the network's output is from the true target. Squaring harms the larger errors and 1/2 simplifies the derivative because $\frac{\partial E}{\partial y} = \hat{y} - y$, which is used in backpropagation.

Verification Table:

Quantity	Manual Calculation	Cell Output	Match?
$a^{(1)}$	$[1.8432\ 1.3410\ 0.6851\ -2.9650]^T$	$[1.8434\ 1.3413\ 0.8851\ -2.96$[illegible]$]$	✓
$a^{(2)}$	-0.9782	-0.9782	✓
$z^{(1)}$	$[0.9511\ 0.672\ 0.709\ -0.9947]^T$	$[0.9511\ 0.672\ 0.709\ -0.9947]$	✓
$\hat{y}$	-0.9782	-0.9782	✓
E	2.8279	2.8278	✓

Output layer gradient derivation:

$E = \frac{1}{2}(a^{(1)} - y)^2$

Let $u = a^{(1)} - y$, so $E = \frac{1}{2}u^2$

$$\frac{\partial E}{\partial a^{(1)}} = \frac{\partial E}{\partial u} \cdot \frac{\partial u}{\partial a^{(2)}} = \frac{1}{2} \cdot 2u \cdot 1 = u = a^{(2)} - y = \hat{y} - y$$

$$\frac{\partial E}{\partial a^{(2)}} = \hat{y} - y = -0.9782 - 1.400 = -2.3782$$

This gradient shows how sensitive the loss is to changes to the output $a^{(2)}$. A positive values means the output is too high and negative values means the output is lower than the target.

Gradient for $W^{(2)}$

$$\frac{\partial E}{\partial w_{ij}^{(2)}} = \frac{\partial E}{\partial a^{(2)}} \cdot \frac{\partial a^{(2)}}{\partial w_{ij}^{(2)}} \qquad \frac{\partial a^{(2)}}{\partial w_{ij}^{(2)}} = z_j^{(1)}$$

$$= (\hat{y} - y) \cdot z_j^{(1)}$$

$$= (\hat{y} - y) \cdot (z^{(1)})^T$$

$$= (-0.9782 - 1.400) \cdot [0.9511, 0.872, 0.709, -0.9947]^T$$

$$= (-2.3782)\, [0.9511, 0.872, 0.709, -0.9947]^T$$

$$= [-2.2619 \;\; -2.0738 \;\; -1.6861 \;\; 2.3656]$$

Gradient for $b^{(2)}$:

$$\frac{\partial E}{\partial b^{(2)}} = \frac{\partial E}{\partial a^{(2)}} \cdot \frac{\partial a^{(2)}}{\partial b^{(2)}} = (\hat{y} - y) \cdot 1 = \hat{y} - y$$

$$= (-0.9782 - 1.400)$$

$$= -2.3782$$

Error signal at hidden layer:

$$\delta^{(1)} = \frac{\partial E}{\partial z^{(1)}} = \frac{\partial E}{\partial a^{(1)}} \cdot \frac{\partial a^{(1)}}{\partial z^{(1)}}$$

$$\frac{\partial a^{(2)}}{\partial z^{(1)}} = W^{(2)}$$

$$\delta^{(1)} = (W^{(2)})^T \cdot \frac{\partial E}{\partial a^{(2)}}$$

$$\delta_j^{(1)} = W_{1j}^{(2)} \cdot (\hat{y} - y)$$

$$\delta_j^{(1)} = [-0.7209 \ \ -0.7083 \ \ -0.8148 \ \ -0.9076](-0.9782 - 1.400)$$

$$= [-0.7209 \ \ -0.7083 \ \ -0.8148 \ \ -0.9076](-2.3782)$$

$$= [1.7144 \ \ 1.6845 \ \ 1.9378 \ \ 2.1585]$$

The transpose $(W^{(2)})^T$ is used so the dimensions line up correctly during the calculation. Each $\delta_j^{(1)}$ represents how much the j-th hidden neuron influenced the output error.

Activation function derivative: Derive $h'(a^{(1)})$ for tanh

$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ Let $u = e^z - e^{-z}$ and $v = e^z + e^{-z}$

$\frac{d}{dz}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$ $u' = e^z + e^{-z}$ $v' = e^z - e^{-z}$

$$h'(z) = \frac{(e^z + e^{-z})(e^z + e^{-z}) - (e^z - e^{-z})(e^z - e^{-z})}{(e^z + e^{-z})^2}$$

$$= \frac{4}{(e^z + e^{-z})^2}$$

$$h'(z) = 1 - \tanh^2(z)$$

$z_i^{(1)} = \tanh(a_i^{(1)})$:

$$h'(a_i^{(1)}) = 1 - (z_i^{(1)})^2$$

$h'(a_1^{(1)}) = 1 - (0.9511)^2 = 0.0954$
$h'(a_2^{(1)}) = 1 - (0.872)^2 = 0.2396$
$h'(a_3^{(1)}) = 1 - (0.709)^2 = 0.4973$
$h'(a_4^{(1)}) = 1 - (-0.9947)^2 = 0.0106$

$h'(a^{(1)}) = [0.0954 \quad 0.2396 \quad 0.4973 \quad 0.0106]$

Gradient with respect to pre-activations: Compute $\frac{\partial E}{\partial a^{(1)}}$

$$\frac{\partial E}{\partial a_i^{(1)}} = \frac{\partial E}{\partial z_i^{(1)}} \cdot \frac{\partial z_i^{(1)}}{\partial a_i^{(1)}} = \delta_i^{(1)} \cdot h'(a_i^{(1)})$$

$$\frac{\partial E}{\partial a^{(1)}} = \delta^{(1)} \odot h'(a^{(1)})$$

$$= [1.7144 \cdot 0.0954 \quad 1.6845 \cdot 0.2396 \quad 1.9378 \cdot 0.4973 \quad 2.1565 \cdot 0.0106]$$

$$= [0.1636 \quad 0.4036 \quad 0.9637 \quad 0.0229]^T$$

The Hadamard product $\odot$ represents element-wise multiplication rather than matrix multiplication. The activation $h'(a^{(1)})$ scales how much of the error signal is passed back through each hidden neuron.

Gradient for $W^{(1)}$: Derive $\frac{\partial E}{\partial W^{(1)}}$ using the chain rule

$$\frac{\partial E}{\partial w_{ij}^{(1)}} = \frac{\partial E}{\partial a_i^{(1)}} \cdot \frac{\partial a_i^{(1)}}{\partial w_{ij}^{(1)}} \qquad \frac{\partial a_i^{(1)}}{\partial w_{ij}^{(1)}} = x_j$$

$$\frac{\partial E}{\partial w_{ij}^{(1)}} = \frac{\partial E}{\partial a_i^{(1)}} \cdot x_j$$

$$\frac{\partial E}{\partial W^{(1)}} = \frac{\partial E}{\partial a^{(1)}} \cdot x^T$$

$$= \begin{bmatrix} 0.1636 \\ 0.4036 \\ 0.9637 \\ 0.0229 \end{bmatrix} \cdot \begin{bmatrix} 5.1 & 3.5 & 0.2 \end{bmatrix} = \begin{bmatrix} 0.8344 & 0.5726 & 0.0327 \\ 2.0584 & 1.4126 & 0.0807 \\ 4.9149 & 3.3730 & 0.1927 \\ 0.1168 & 0.0802 & 0.00458 \end{bmatrix}$$

Gradient for $b^{(1)}$: Derive $\frac{\partial E}{\partial b^{(1)}}$

$$\frac{\partial E}{\partial b_i^{(1)}} = \frac{\partial E}{\partial a_i^{(1)}} \cdot \frac{\partial a_i^{(1)}}{\partial b_i^{(1)}} = \frac{\partial E}{\partial a_i^{(1)}} \cdot 1$$

$$\frac{\partial E}{\partial b^{(1)}} = \frac{\partial E}{\partial a^{(1)}}$$

$$= \begin{bmatrix} 0.1636 \\ 0.4036 \\ 0.9637 \\ 0.0229 \end{bmatrix}$$

Complete chain rule example:

$$\frac{\partial E}{\partial w_{12}^{(1)}} = \frac{\partial E}{\partial a^{(2)}} \cdot \frac{\partial a^{(2)}}{\partial z_1^{(1)}} \cdot \frac{\partial z_1^{(1)}}{\partial a_1^{(1)}} \cdot \frac{\partial a_1^{(1)}}{\partial w_{12}^{(1)}}$$

$$\frac{\partial E}{\partial w_{12}^{(1)}} = (-2.3782)(-0.7209)(0.0954)(3.5)$$

$$= 0.57245$$

Verification table:

Gradient	Manual Derivation	Cell Output	Match?
$\frac{\partial E}{\partial a^{(1)}}$	-2.3782	-2.3782	✓
$\frac{\partial E}{\partial w_{11}^{(2)}}$	-2.2619	-2.2619	✓
$\frac{\partial E}{\partial w_{12}^{(2)}}$	-2.0738	-2.0737	✓
$\frac{\partial E}{\partial w_{13}^{(2)}}$	-1.6861	-1.6861	✓
$\frac{\partial E}{\partial w_{14}^{(1)}}$	2.3656	2.3656	✓
$\delta_1^{(1)}$	1.7144	1.7145	✓
$\delta_2^{(1)}$	1.6845	1.6845	✓
$\delta_3^{(1)}$	1.9378	1.9376	✓
$\delta_4^{(1)}$	2.1585	2.1585	✓
$h'(a_1^{(1)})$	0.0954	0.0954	✓
$h'(a_2^{(1)})$	0.2396	0.2397	✓
$h'(a_3^{(1)})$	0.4973	0.4673	✓
$h'(a_4^{(1)})$	0.0106	0.0106	✓
$\frac{\partial E}{\partial w_{12}^{(1)}}$	0.57245	0.5723	✓